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ELECTRONIC SUPPLEMENTARY MATERIAL

Modeling spatially-resolved characterization factors for eutrophication potential in life cycle assessment

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1. Data and Methods

This research uses likelihood data (Table S1 and Table S2) to calculate an aggregated value of each target polygon. Table S1 lists the likelihood data categories of P (for freshwater eutrophication) while Table S2 lists likelihood data categories of N (for marine eutrophication).

Table S1 Likelihood data for phosphorus used from the National Inventory of Phosphorus (Sabo and Clark 2019)

Data Item	Agricultural	Nonagricultural	General
Urban Fertilizer (kg P yr^{-1})		X	X
Agricultural Fertilizer (kg P yr^{-1})	X		X

Table S2 Likelihood data for nitrogen used from the National Inventory of Nitrogen (Clark and Sabo 2018)

	N			NOx			NHx		
Data Item	Agric	Nonagric	General	Agric	Nonagric	General	Agric	Nonagric	General
Agricultural Fertilizer (kg N yr^{-1})	X		X	X		X			
Total atmospheric N deposition onto agricultural land (kg N yr^{-1})	X		X	X		X			
Urban Fertilizer (kg N yr^{-1})		X	X		X	X		X	X
Agricultural NHx emissions (kg N yr^{-1})							X		X
Fossil Fuel Combustion: transportation emissions (kg N yr^{-1})		X	X		X	X			
Fossil Fuel Combustion: utility and industry emissions (kg N yr^{-1})		X	X		X	X			

2. Inventory data

Fig. S1illustrates the national phosphorus and nitrogen inventory at the HUC 8 watershed level used for the case application (see Section 3).

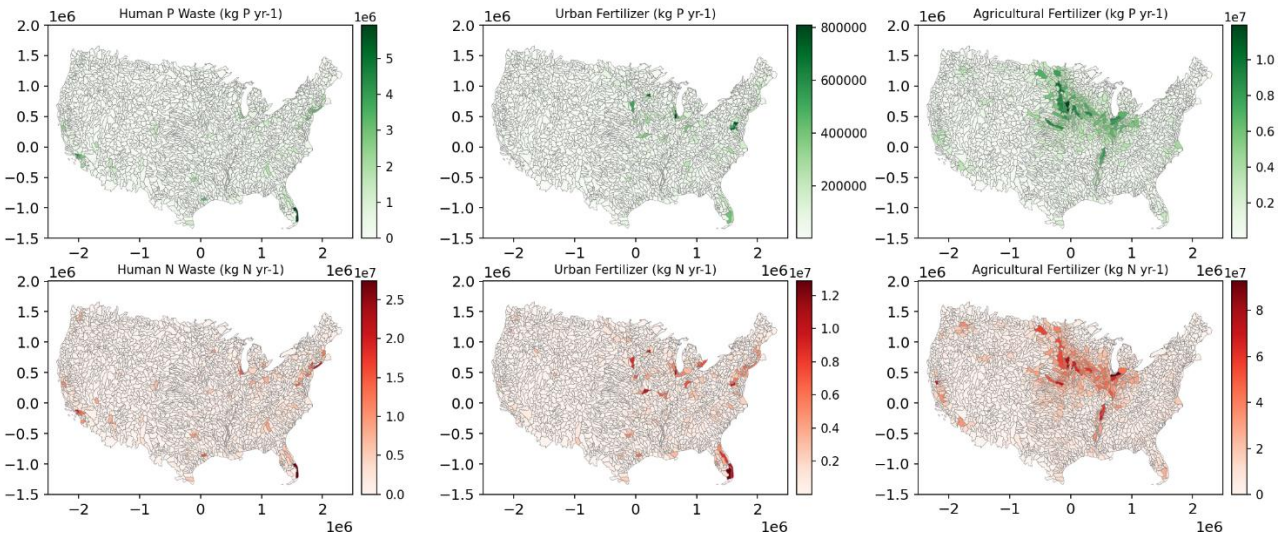

Fig. S1 Inventory for P and N emissions selected for inclusion in the application example

3. Case application

Fig. S2 and Fig. S3 expand upon Fig. 7 and Fig. 8, respectively, in the main text. The maps in the main text are drawn at the state-level, whereas Figs. S2 and S3 are drawn at the county-level.

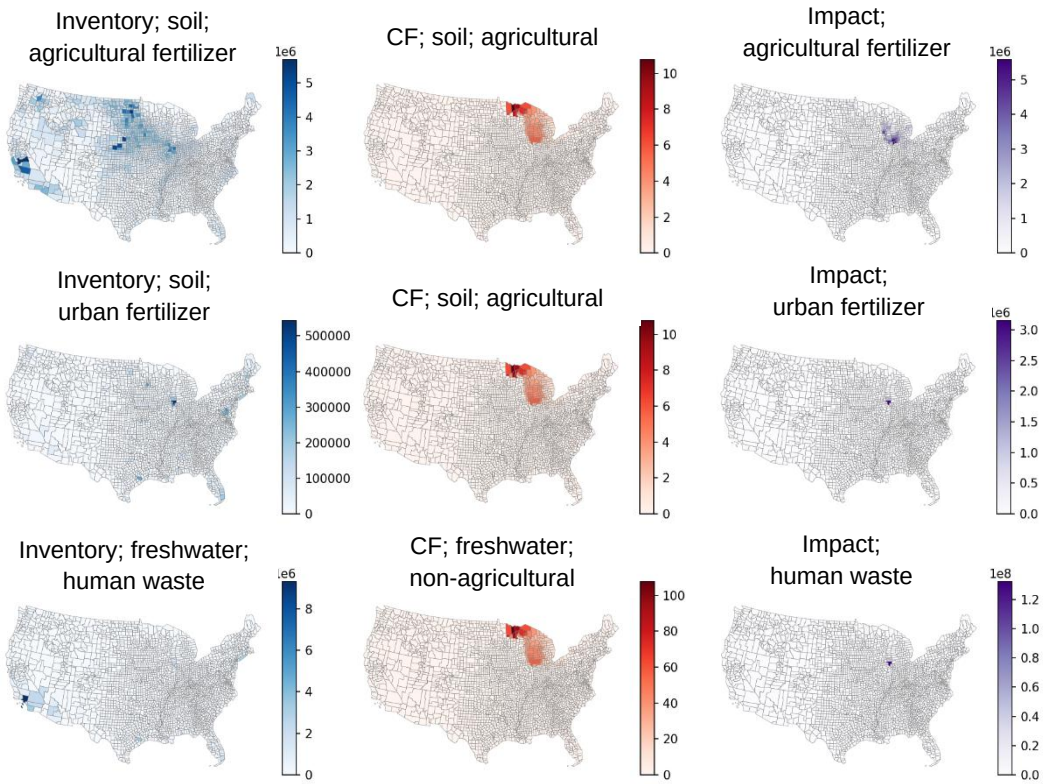

Fig. S2 Case application for P (freshwater eutrophication) at the county level: agricultural fertilizer (top row); urban fertilizer (middle), and human waste (bottom). Columns represent the national inventory translated to county (left column, kg P/yr), the corresponding characterization factor (center column, CF, yr) and the impact (right column, kg P).

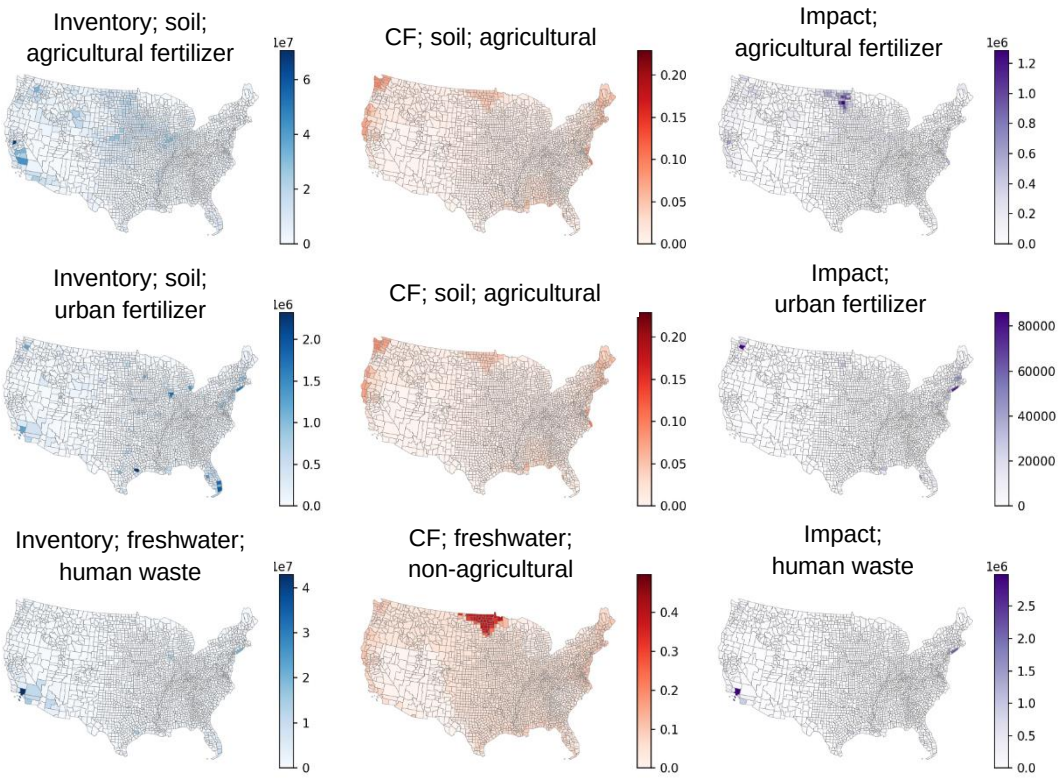

Fig. S3 Case application for N (marine eutrophication) at the county level: agricultural fertilizer (top row); urban fertilizer (middle), and human N waste (bottom). Columns represent the national inventory translated to county (left column, kg N/yr), the corresponding characterization factor (center column, CF, yr) and the impact (right column, kg N).

4. References

Clark C, Sabo R (2018) National Inventory of Reactive Nitrogen. U.S. Environmental Protection Agency: Environmental Dataset Gateway. https://doi.org/10.23719/1418976

Sabo R, Clark C (2019) National Inventory of Phosphorus (v1, 8.30.19). U.S. Environmental Protection Agency. https://doi.org/10.23719/1504278